Parametric Survival: Weibull

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Introduction

The Weibull distribution is the most flexible parametric survival family that has a monotone hazard. Hazard is increasing when the shape parameter $k > 1$, constant (exponential) when $k = 1$, and decreasing when $k < 1$. Among the common parametric families it is unique in admitting both proportional-hazards and accelerated-failure-time interpretations, which makes it the natural starting point for parametric survival analysis whenever PH is plausible. Cleaner extrapolation, smaller standard errors, and explicit hazard shape are the practical pay-offs over the semi-parametric Cox model when its assumptions hold.

Prerequisites

A working understanding of the Weibull distribution, survival analysis, and the distinction between proportional-hazards and accelerated-failure-time parameterisations.

Theory

The Weibull survival function is

$$S(t) = \exp\left(-(t/\lambda)^k\right),$$

with hazard $h(t) = (k/\lambda)(t/\lambda)^{k-1}$. The hazard is monotone: increasing for $k > 1$ (positive ageing), decreasing for $k < 1$ (early-failure regimes), and constant for $k = 1$ (exponential).

Two parameterisations coexist:

- **Accelerated-failure-time (AFT)** in `survreg(dist = "weibull")`: $\log T = X\beta + \sigma W$ with W a standard extreme-value variate; the scale is $\sigma = 1/k$.
- **Shape-scale** in `flexsurv::flexsurvreg(dist = "weibull")`: explicit shape k and scale λ, with PH-style covariate effects via interactions on λ.

The conversion between PH and AFT forms is $\beta_{\mathrm{PH}} = -k \cdot \beta_{\mathrm{AFT}}$.

Assumptions

A monotone hazard. The standard diagnostic is the log–log survival plot: $\log[-\log \hat{S}(t)]$ against $\log t$ should be approximately linear with slope k if the Weibull holds.

R Implementation

```
library(survival); library(flexsurv)

data(lung)

fit_survreg <- survreg(Surv(time, status) ~ age + sex, data = lung,
                       dist = "weibull")
summary(fit_survreg)
```

```r
# flexsurvreg gives shape and scale directly
fit_flex <- flexsurvreg(Surv(time, status) ~ age + sex, data = lung,
                        dist = "weibull")
fit_flex
```

Output & Results

`survreg()` returns AFT coefficients (interpret as multiplicative effects on event time, $\exp(-\beta)$) plus the scale parameter. `flexsurvreg()` returns shape, scale, and PH-style hazard ratios directly — easier to communicate to a Cox-fluent audience. Both yield the same likelihood and the same fitted survival.

Interpretation

A reporting sentence: "A Weibull AFT model estimated shape parameter $\hat{k} = 1.31$ (95 % CI 1.10 to 1.55), indicating an increasing hazard over follow-up; each additional year of age accelerated the event by approximately 2 %, equivalent to a hazard ratio of 1.025 per year on the PH scale." When reporting, choose AFT or PH consistently with your audience: clinical readers expect HRs, while time-to-event-cost analyses prefer time-acceleration ratios.

Practical Tips

- Use the log–log survival plot (`plot(survfit(...), fun = "cloglog")`) as the main diagnostic for Weibull suitability; clear non-linearity rules it out.
- For more flexible monotone hazards, compare AIC against generalised gamma and Gompertz; for non-monotone, switch to log-normal or Royston-Parmar splines.
- Convert between PH and AFT carefully: $\beta_{\mathrm{PH}} = -k \cdot \beta_{\mathrm{AFT}}$, and remember that AFT confidence intervals are on the log-time scale.
- For small samples, Weibull yields smoother survival estimates than Kaplan-Meier and lower-variance covariate effects than Cox; this is its main practical advantage over Cox.
- `flexsurv` simplifies prediction (`summary(fit, type = "survival", t = ...)`) and supports time-dependent shape parameters via spline extensions.
- Check Weibull plus Cox HR estimates; they should be close under PH. Material disagreement points to an assumption failure that should be investigated.

R Packages Used

`survival` for `survreg()` with `dist = "weibull"`; `flexsurv` for `flexsurvreg()` with shape/scale output and richer prediction; `eha` for additional parametric extensions; `rstpm2` for Royston-Parmar spline alternatives when monotone hazards are too restrictive.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Survival primer: KM, log-rank, Cox PH
- Time-varying covariates and immortal-time bias

- Competing risks and multistate models
- Survival ML (random survival forest, DeepSurv)
- Time-dependent Brier, IPA, external validation